

JEE MAIN REVISION MASTER: SIMPLE HARMONIC MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	SHM Kinematics & Energy Relations	7 PYQs	Oscillations
Standard	Time Period of Simple Pendulum & Spring-Mass Systems	2 PYQs	Oscillation Systems

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
SHM Kinematics & Energy Relations	$v = \omega \sqrt{A^2 - x^2}$, $a = -\omega^2 x$, $E_{\text{total}} = \frac{1}{2} m \omega^2 A^2 = 2\pi^2 m f^2 A^2$
Time Period of Simple Pendulum & Spring-Mass Systems	$T_{\text{pendulum}} = 2\pi \sqrt{\frac{L}{g}}$, $T_{\text{spring}} = 2\pi \sqrt{\frac{m}{k}}$, $k \times L = \text{constant}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 SHM Kinematics & Energy Relations (Moderate Yield — 7 PYQs)

Core Principle: Kinematics of SHM: $x = A \sin \omega t$, $v = \omega \sqrt{A^2 - x^2}$, $a = -\omega^2 x$. At mean position ($x=0$): $v = v_{\text{max}} = A\omega$, $a = 0$, K is maximum. At extremes ($x=\pm A$): $v = 0$, $a = a_{\text{max}} = \omega^2 A$, U is maximum. Total energy $E = \frac{1}{2} m \omega^2 A^2$ is constant at all positions.

■ **Caution / JEE Pitfall:** Kinetic and potential energy oscillate with frequency $2f$ (twice the frequency of displacement x)!

■ **Speed Trick:** At what displacement is $K = U$? When $x = \frac{A}{\sqrt{2}}$. At what displacement is $K = 3U$? When $x = (A / 2)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A mass of 1 kg is executing SHM. Its displacement is given by $x=6.0\cos(100t+\pi/4)$ cm. What is the maximum kinetic energy?..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The variations of kinetic energy $K(x)$, potential energy $U(x)$ and total energy as a function of displacement of a particle in SHM is as ..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "For a particle executing simple harmonic motion (SHM), at its mean position..." [Ans: N/A]

3.2 Time Period of Simple Pendulum & Spring-Mass Systems (Standard Yield — 2 PYQs)

Core Principle: Time period of simple pendulum: $T = 2\pi\sqrt{(L/g)}$. A seconds pendulum has period $T = 2$ s and length $L \approx 1$ m. Spring-mass system: $T = 2\pi\sqrt{(m/k)}$. If a spring of constant k is cut into n equal parts, each part has stiffness $k' = n k$.

■ **Caution / JEE Pitfall:** Time period of a spring-mass system is independent of gravity g ; taking a spring-mass oscillator to the Moon does not change its frequency!

■ **Speed Trick:** If the length of a pendulum increases by 2%, its time period increases by 1% ($T \propto L^{1/2}$). It loses 432 seconds per day.

Representative Source PYQs:

- **[2020 Shift] Q#3 (MCQ):** "A tray of mass 12kg is supported by two identical springs as shown in figure. When the tray is pressed down slightly and then released, it..." **[Ans: N/A]**
- **[2017 Shift] Q#1 (MCQ):** "Two simple pendulums A and B are made to oscillate simultaneously and it is found that A completes 10 oscillations in 20 sec and B c..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A mass of 1 kg is executing SHM. Its displacement is given by $x = 6.0\cos(100t + \pi/4)$ cm. What is the maximum kinetic energy?

- (A) 3 J
- (B) 6 J
- (C) 9 J
- (D) 18 J

Correct Answer: None

Solution Analysis: For SHM, $x = A\cos(\omega t + \phi)$ Given displacement is $x = 6.0\cos(100t + \pi/4)$ cm So the amplitude is $A = 6.0\text{cm} = 0.06\text{m}$ and angular frequency is $\omega = 100\text{rad s}^{-1}$ In SHM, maximum kinetic energy is equal to the total mechanical energy: $K_{(\max)} = (1/2)m\omega^2 A^2$ Substitute the values: $K_{(\max)} = (1/2)(1)(100)^2(0.06)^2$ $K_{(\max)} = (1/2) \times 10000 \times 0.0036$ $K_{(\max)} = 18\text{J}$ So the maximum kinetic energy is 18J Correct option: Option D.

Benchmark Problem 2: KCET 2020 (Q#3)

A tray of mass 12kg is supported by two identical springs as shown in figure. When the tray is pressed down slightly and then released, it executes SHM with a time period of 1.5s. The spring constant of each spring is

- (A) 50 Nm^{-1}
- (B) 0
- (C) 105 Nm^{-1}
- (D) ∞

Correct Answer: None

Solution Analysis: Mass of tray, $m = 12\text{kg}$ Time period, $T = 1.5\text{s}$ If k be the spring constant of each spring, then $k_1 = k_2 = k$ since, springs are connected in parallel, hence $k_{(\text{ba})} = k + k$ Time period, $T = 2\pi\sqrt{\frac{m}{k_{(\text{net})}}} = 2\pi\sqrt{\frac{m}{k_1 + k_2}} \Rightarrow 1.5 = 2\pi\sqrt{\frac{12}{k + k}} \Rightarrow 1.5 = 2\pi\sqrt{\frac{12}{2k}} \Rightarrow 1.5 = 2\pi\sqrt{\frac{6}{k}}$ Squaring both side, we have
$$2.25 = 4\pi^2 \cdot \frac{6}{k} \Rightarrow k = \frac{4\pi^2 \times 6}{2.25} \approx 105.17\text{Nm}^{-1} \approx 105\text{Nm}^{-1}$$